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Needle loader devices and related systems and methods

Abstract

Implementations of a needle loader and related methods are provided. The needle loader comprises a housing, a movable arm disposed within the housing, and a needle mount defined on the movable arm to releasably attach to an arced suturing needle. The needle loader further includes an arced needle releasably attached to the needle mount. The needle can be held in place on the needle mount by way of an interference fit. The needle mount can be defined on a surface of the movable arm that faces an arced needle track of a suturing device when loading an arced suturing needle in the needle track of the suturing device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present patent application claims the benefit of priority to U.S. Provisional Patent Application No. 63/197,786, filed Jun. 7, 2021. The content of the aforementioned patent application is incorporated by reference herein in its entirety for all purposes.

TECHNICAL FIELD

(1) Aspects of the present disclosure relate to devices, systems, and methods for performing a suturing procedure. For example, aspects of the present disclosure relate to needle loader devices including, but not limited to, for example, devices configured to receive a suturing device to load a suturing needle therein in support of remote surgical, diagnostic, therapeutic, and other treatment procedures. Further aspects of the disclosure relate to methods of operating such devices.

INTRODUCTION

(2) Sutures are used in a variety of surgical and other applications, such as closing ruptured or incised tissue, soft tissue attachment, attachment of grafts, etc. Additionally, sutures may have other medical and/or non-medical uses. Conventionally, suturing is accomplished by penetrating tissue with the sharpened tip of a suturing needle that has a thread of suturing material attached to the opposite blunt end of the needle. The needle is then pulled through the tissue, causing the attached thread of suturing material to follow the path of the needle. Typically, a knot is tied at the trailing end of the thread to anchor the first stitch. This action is performed repetitively with application of tension to the needle to pull a length of the thread through the tissue using subsequent stitches until the tissue is sutured as desired with one or more stitches.

(3) While the above-described suturing process can be performed manually, automated suturing systems also exist. Such systems can include a needle driver device that has an open, C-shaped portion into which tissue segments are introduced. The C-shaped portion defines two arms, each with an entry/exit point for a curved needle. The curved needle is driven around a track (generally following the C-shaped portion) and across the opening in the C-shaped portion to draw a thread of suturing material into the needle driver device through the tissue segments, similar to the manual suturing process discussed above. It is desirable to provide needle driver devices that occupy a minimal amount of space relative to a size (e.g., gauge and/or radius) of the needle. Such tools are useful in space-limited applications, such as in the case of minimally invasive surgery, for example laparoscopic surgery including both manual and computer-assisted surgery.

(4) A need exists to provide reliable devices and techniques to load a suturing needle into these types of devices. While some approaches have been proposed and used, there is considerable room for improvement to provide safe and reliable devices.

SUMMARY

(5) Example embodiments of the present disclosure may solve one or more of the above-mentioned problems and/or may demonstrate one or more of the above-mentioned desirable features. Other features and/or advantages may become apparent from the description that follows.

(6) In accordance with some aspects of the present disclosure, a needle loader is provided. The needle loader includes a housing, a movable arm or assembly disposed within the housing, and a needle mount, which can optionally be defined on the movable arm or be located nearby, to releasably attach to an arced suturing needle.

- (7) In accordance with at least another aspect of the present disclosure, a needle loader can be provided that additionally or alternatively includes an arced needle releasably attached to the needle mount. The needle can be held in place attached to the needle mount by way of an interference fit or other suitable technique.
- (8) In accordance with at least another aspect of the present disclosure, a needle loader can be provided wherein the needle mount is defined on a surface of the movable arm that faces an arced needle track of a suturing device when loading an arced suturing needle in the needle track of the suturing device.
- (9) In accordance with further aspects of the present disclosure, a needle loader can be provided wherein the needle loader is configured to removably engage with a distal portion of a suturing device, wherein the suturing device typically includes an arced needle track.
- (10) In accordance with further aspects of the present disclosure, a needle loader can be provided wherein a movable arm is movable from a first position wherein the needle mount is spaced apart from a distal portion of the suturing device to a second position wherein the needle mount is adjacent an arced needle track of the suturing device. In some implementations, the movable arm can deform or flex as the needle mount approaches the second position. For example, the movable arm can flex against a distal portion of a suturing device as the needle mount approaches the second position.
- (11) In accordance with another aspect of the present disclosure, a needle loader can be provided wherein a needle mount of the needle loader includes one or more inclined surfaces that cause an arced needle mounted on the needle mount to approach a distal portion of a suturing device as the needle mount moves from a first position wherein the needle mount is spaced apart from the distal portion of the suturing device to a second position wherein the needle mount is adjacent to an arced needle track of the suturing device. For example, if desired, the needle mount and/or inclined surface can be configured to permit the inclined surface to slide over a boss as the needle mount moves from the first position to the second position. This can cause an arced needle mounted on the needle mount to approach a distal portion of a suturing device.
- (12) In accordance with at least another aspect of the present disclosure, a needle loader can be provided wherein a needle mount or movable arm can lock in position after it has reached a second, or displaced position. For example, the needle mount or movable arm can lock in position after it has reached a second position by action of a locking element. A spring-loaded barb or pawl coupled to the movable arm can pass over a boss defined on the housing of the needle loader. In another implementation, the needle mount or movable arm can be locked in position after it has reached the second position by a surface defined on the movable arm passing over a pawl defined in the housing of the needle loader. In some implementations, the needle mount movable arm can define a locking surface thereon to engage a retention surface defined by the housing when the movable arm has reached the second position. The engagement of the locking surface and the retention surface can prevent the movable arm from returning to the first position.
- (13) In accordance with another aspect of the present disclosure, a needle loader can be provided that includes a spring that urges against motion of a movable arm as the movable arm moves from a first position to a second position. For example, such a spring can comprise a leaf spring, a compression spring or a tension spring. In some implementations, the movable arm can be coupled to the spring. In some implementations, the movable arm can include a first end coupled to a rotatable hub and a second free end including the needle mount. In some implementations, the spring can also be coupled to the rotatable hub. In some implementations, the spring can comprise a deflectable hinge. In some implementations, the spring can be integrally formed with the needle mount.
- (14) In accordance with still another aspect of the present disclosure, a needle loader can be provided having a movable arm that includes a cam surface that is configured to slide over a distal portion of a suturing device to be loaded with an arced suturing needle as the movable arm

approaches the second position.

(15) In accordance with yet another aspect of the present disclosure, a needle loader can be provided wherein movement of a movable arm of the needle loader from a first position to a second position can activate an indicator confirming that the movable arm has reached the second position. The indicator can confirm that the movable arm has reached the second position by emitting a sound, a tactile signal, and/or by displaying one or more visual indicia. The indicator can confirm that the movable arm has reached the second position by moving from a first indicator position to a second indicator position. The movable arm can include a boss that slides in a guide track formed into an inner surface of the housing.

(16) In accordance with additional aspects of the present disclosure, a needle loader can be provided that comprises a protrusion that extends into a path of a suturing device to be loaded with a needle from the needle loader. The protrusion can cause an arced needle track of the suturing device to be exposed as the suturing device is advanced along the path.

(17) In accordance with additional aspects of the present disclosure, a needle loader can be provided wherein a housing of the needle loader further defines an opening therein for receiving the suturing device. The opening can at least in part define the path to insert and load the needle into the suturing device.

(18) In accordance with additional aspects of the present disclosure, a needle loader can be provided that can receive a suturing device that includes a retractable cover to cover the needle track. A boss, protrusion, or other feature can retract the retractable cover of the suturing device to expose an arced needle track or other feature to receive an arced suturing needle from the needle loader. In some implementations, the protrusion or other feature can be coupled to the needle mount. In other implementations, the protrusion can be coupled to the housing.

(19) In accordance with additional aspects of the present disclosure, a needle loader can be provided having a slidable arm, or a movable arm that can be rotatable about at least one pivot point between a first position and a second position to load an arced needle into a suturing device. In some implementations, the movable arm can comprise a four-bar linkage. The four-bar linkage of the movable arm can be rotatable about two pivot points. In some implementations, the four-bar linkage can comprise two generally parallel pivotable arms, wherein each generally parallel pivotable arm includes a first end and a second end, and further wherein a respective first end of each of the parallel pivotable arms can be rotatably coupled to the housing. In some implementations, a respective second end of each of the parallel pivotable arms can be coupled to the needle mount by a hinge.

(20) In accordance with additional aspects of the present disclosure, a needle loader can be provided that comprises a needle mount that in turn comprises a first surface defining a boss or retainer thereon to receive an arced needle. The needle mount can further include a second surface that faces the first surface of the needle mount. The first surface of the needle mount and the second surface of the needle mount can cooperate to define a gap to receive a distal portion of the suturing device. At least one of the first surface of the needle mount and the second surface of the needle mount can comprise a ramped surface to guide an arced needle track of a suturing device toward an arced needle mounted on the needle loader as the suturing device is advanced along a path of travel with respect to the needle loader.

(21) In accordance with additional aspects of the present disclosure, a needle loader can be provided comprising a needle mount that can be slidably disposed within a housing. For example, a needle mount can be slidably movable from a first position wherein the needle mount is not engaged with a distal portion of a suturing device to a second position wherein the needle mount is engaged with the distal portion of the suturing device.

(22) In accordance with additional aspects of the present disclosure, a needle loader can be provided that comprises a suture compartment to receive a length of suture coupled to an arced needle. The length of suture can be, for example, between about six and about 24 inches in length.

In some implementations, the suture compartment can be coupled to a movable needle mount. If desired, the suture compartment can be integrally formed with the needle mount. In some implementations, the suture compartment can be configured to slide with or without the needle mount within the housing. If desired, the suture compartment can comprise at least one boss that is slidably received in a respective track formed into the housing. In some implementations, the suture compartment can be oblong in shape. In some implementations, the suture compartment can be defined by a peripheral curved wall. The peripheral curved wall is circular in shape, elliptical in shape, or a combination of different shapes. In some implementations, the suture compartment can be located in a first compartment defined by the housing that is distinct and at least partially separated from a second compartment that houses the movable arm. The first compartment and the second compartment can be joined by a groove, and a suture coupled to an arced suturing needle that is coupled to the needle mount disposed in the second compartment can extend through the groove to a predetermined length of suture in the first compartment.

(23) In accordance with additional aspects of the present disclosure, the housing of the needle loader can define one or more gripping surfaces to facilitate gripping by a user to hold the needle loader.

(24) Additional objects, features, and/or advantages will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the present disclosure and/or claims. At least some of these objects and advantages may be realized and attained by the elements and combinations particularly pointed out in the appended claims.

(25) It is to be understood that both the foregoing general description and the following detailed description are illustrative and explanatory only and are not restrictive of the claims; rather the claims should be entitled to their full breadth of scope, including equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiments of the present teachings and together with the description explain certain principles and operation. In the drawings,

(2) FIGS. 1A-1H depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(3) FIGS. 2A-4E depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects or variations thereof.

(4) FIGS. 5A-5O depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(5) FIGS. 6A-6E depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(6) FIGS. 7A-7N depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(7) FIGS. 8A-8C depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(8) FIGS. 9A-9P depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(9) FIGS. 10A-11D depict views of a needle loader device in accordance with some embodiments of the present disclosure, or aspects thereof.

(10) FIGS. 11E-11F depict views of a needle loader device in accordance with some embodiments

of the present disclosure, or aspects thereof.

(11) FIG. 12 is a perspective view of a manipulator system according to an illustrative embodiment of the disclosure.

(12) FIG. 13 is a partial schematic view of an embodiment of a manipulator system having a manipulator arm with two instruments in an installed position according to the present disclosure.

DETAILED DESCRIPTION

(13) The present disclosure provides various embodiments of needle loader devices, systems, and methods. Needle loader devices according to various embodiments of the present disclosure include features that facilitate the loading of a suturing device with an arced suturing needle.

(14) In accordance with some aspects of the present disclosure, a needle loader is provided. The needle loader includes a housing, a movable arm disposed within the housing, and a needle mount defined on the movable arm to releasably attach to an arced suturing needle.

(15) For purposes of illustration, and not limitation, a representative implementation of a needle loader is presented in FIGS. 1A-1H. Further embodiments of needle loaders, or aspects thereof, are illustrated in FIGS. 2A-11F.

(16) The needle loader **100** of FIGS. 1A-1H includes a housing **110**, a movable arm **120** disposed within the housing **110**, and a needle mount **130** defined on the movable arm **120** to releasably attach or releasably couple to an arced suturing needle **135**. FIGS. 1A-1B illustrate a right-side elevational view and left side elevational view, respectively, of needle loader **100**. FIGS. 1C-1E illustrate interior views of the needle loader **100**. As described in further detail below, the housing **110** is configured to receive a portion of a suturing device **10** such that insertion of the suturing device **10** into the housing **110** causes the needle mount **130** to move from a first position to a second position and transfer (e.g., load) the suturing needle **135** into engagement with the suturing device **10**.

(17) Referencing FIG. 1F, with respect to embodiment **100**, needle loading is accomplished by way of rotary motion of a movable arm, or arm assembly **120**, about a boss or axle **111**. Specifically, forward motion of a suturing device **10** into an opening **108** of a housing **110** of the device urges against a portion (e.g., **126a**) of the arm or arm assembly **120** to cause the arm or arm assembly **120** to rotate clockwise (with regard to FIG. 1F) so that a mount **130** including the needle **135** mounted thereon moves from a first location wherein it is out of alignment with a path of travel of the suturing device **10** to a second location wherein the mount **130** and needle **135** engage with the suturing device **10**, such that continued forward motion of the device **10** into passage **108** then causes the needle to align with and be pressed into an arcuate needle track of the suturing device so that the needle **135** becomes engaged with the needle track. Optionally, at the same time, the continued motion of the arm or arm assembly **120** about the boss or axle **111** can cause a lock **142** to engage, to prevent backward rotation, or movement of the arm or arm assembly **120** about the boss or axle **111**. As such, once this engagement has occurred, the needle **135** is trapped by the needle track of the suturing device **10**, and the device can be withdrawn.

(18) It will be appreciated that a number of the embodiments disclosed herein provide a tunnel or passageway (e.g., **108**) into which a suturing device **10** is inserted. This passageway, in cooperation with the supporting housing (e.g., **110**), act to isolate the needle **135** from a user that is loading the needle onto the suturing device **10** during the loading process. And, optionally, the needle loader **100** can be further configured to permit the suturing device **10** to close over the needle track containing the needle **135** as the suturing device **10** is withdrawn from the needle loader, still further isolating the needle from inadvertent contact. Furthermore, the needle mount **130** holds the needle **135** in the needle loader **100**, and mounts the needle **135** to the suturing device **10** in an orientation whereby the sharp end and the second end of the needle **135** are wholly contained within the needle track of the suturing device such that, when the suturing device **10** is extracted from the needle loader **100**, only the middle of the body of the needle **135** is exposed in the gap of the suturing device **10**. In some embodiments, optionally, each end of the needle may be fully

covered and surrounded by the needle track and sliding cover of the suturing device **10**. The needle can then be advanced to a starting position within the suturing device **10** by cycling the drive system to advance the needle to a relative position within the needle track of the suturing device **10** wherein the entire needle, including both ends, is situated within the suturing device and the sharp leading end of the suturing device is protected so that it will prevent accidental needle contact.

(19) It will be appreciated that the needle loader **100** containing the needle **135** with suture material coupled thereto can be provided in a sterile kit surrounded by a removable envelope and/or container that is ready to be opened upon use in a procedure. The kit may be a single use device, or may be reprocessed and reloaded with a new/sterilized needle as desired.

(20) The arced needle **135** can be releasably held in place attached to the needle mount **130** by way of an interference fit. The needle mount **130** may include needle engagement portions configured to releasably hold the surface of the arced needle **135**. For example, a flexible member, such as one or more cleats, clips or fingers that can flex and release the needle after delivery can be used. Other suitable techniques can be used, such as a magnetic needle mount **130** with a ferritic needle **135**. Alternatively, a deformable material can be provided as a part of the needle mount **130**, or a relatively high friction or tacky surface. It will be appreciated that these techniques can be used to hold a needle in place in the other implementations described herein.

(21) As depicted, the needle mount **130** is defined on a surface of the movable arm **120** that faces an arced needle track of a suturing device **10** when loading an arced suturing needle **135** in an arcuately shaped needle track of the suturing device **10**. An example of a suturing device suitable for being loaded by needle loader **100** can include, for example, those described in U.S. Pat. No. 7,862,572, which is incorporated by reference herein in its entirety for all purposes.

(22) As depicted, the arm **120** is pivotally mounted to the housing **110**. In the illustrated embodiment, the arm **120** is mounted to a hub **121** defining a passage therethrough that is rotatably received on a boss or axle **111** defined by the housing **110**. As such, the movable arm **120** includes a first end coupled to the rotatable hub **121** and a second free end coupled to the needle mount **130**, which in turn releasably holds the needle **135**.

(23) As depicted in FIG. **1F**, in some embodiments, a spring, such as a leaf spring **160**, may extend outwardly from the hub **121**. The spring **160** is deflectable and can bend. In operation, a lower arm, or boom **126** extends downwardly from the hub **121** (e.g., relative to the position of the arm **120**), and defines a cam surface **126a** thereon.

(24) As mentioned above, to load the needle **135** into the suturing instrument **10**, the suturing instrument **10** is inserted into a cavity, or opening, **108** of the needle loader **100**, and the distal end or end region of the suturing device **10** urges against the cam surface **126a** of the boom **126**. This causes the hub **121** including arm **122** and needle mount **130** with needle **135** to rotate in a clockwise direction (relative to the orientation shown in FIGS. **1C-1E**), resulting in the needle **135** being caused to approach the distal end region of the suturing device **10**. As this is occurring, the leaf spring **160** contacts and begins to flex against boss **115** defined by the housing **110** to provide a countervailing force that resists the forward insertion movement of the suturing device **10** into the housing **110**. In so doing, the spring **160** urges against motion of the movable arm **120** as the movable arm **120** moves from a first position to a second position.

(25) In the event that the movable arm **120** does not move entirely to its second position, such as when the user only partially inserts the suturing device **10** but does not load the needle into the suturing device **10**, the spring **160** also functions as a way for the needle loader **100** to reset itself to the first position. This feature is also set forth in the other embodiments herein.

(26) It will be appreciated that the spring **160** can be configured in various manners. For example, the spring can comprise a leaf spring, a compression spring or a tension spring, among others. As illustrated in the needle loader **100**, the movable arm **120** is integrally formed with the spring **160** in a single molding, but these components can be separated or provided as individual components. Likewise, in other implementations, one or more springs can be provided that are not coupled to a

needle carrying component of the device.

(27) As depicted in FIGS. **1C-1E**, in operation the movable arm **120** of the needle loader **100** can be movable from a first position, such as that in FIG. **1C**, wherein the needle mount **130**, and needle **135** are spaced apart from the distal portion of the suturing device **10** to a second position, such as that in FIG. **1E**, wherein the needle mount is adjacent the arced needle track of the suturing device **10**. In the embodiment of FIGS. **1C-1E**, the arm **120** of the needle loader **100**, and/or the needle mount **130** of the needle loader **100**, can be configured to bend, or flex, as the needle mount **130** with the needle **135** mounted thereon approaches the second position illustrated in FIG. **1E**. Specifically, in the embodiment of FIG. **1E**, the arm **120** can flex against the distal portion of the suturing device **10** as the needle mount **130** approaches the second position, while the leaf spring is also flexing against boss **115**.

(28) As further illustrated in FIGS. **1F** and **1H**, optionally, in some embodiments, the needle mount **130** and movable arm **120** are configured to lock in position after, or as, the needle **135** is loaded into the needle track of the suturing device **10**. This can prevent the needle mount **130** and movable arm **120** from moving from the second position to the first position when, for example, the instrument **10** is removed from the needle loader **100**. For example, a portion of the needle mount **130**, movable arm **120**, and/or spring **160** may include one or more locking surfaces configured to engage one or more retention surfaces of the housing **110** to prevent backwards movement of the assembly to the first position. These or similar locking surface features can also be included in the other embodiments of needle loaders discussed herein.

(29) As illustrated in FIGS. **1F** and **1H**, a boss **142** may be provided on either side of the boom **126** to facilitate locking of the needle mount **130** during or after needle loading. The hub **121**, arm **120**, boom **126**, and spring **160** form a movable assembly that may be locked in position by the bosses **142** engaging respective tabs, barbs, or pawls **112** formed into the housing **110**. As the bosses **142** move relative to the tabs **112**, the bosses **142** may deflect the tabs **112** out of the way, and the tabs **112** may then snap back in place after the bosses **142** pass the tabs **112**, thereby preventing the assembly from rotating backward. At this point, the needle **135** is loaded into the suturing device **10** and the suturing device can be removed from the housing **110**. In some implementations, the needle mount **130**, movable arm **120**, boom **126** and/or spring **160** can define additional or alternative locking surface thereon to engage one or more retention surfaces defined by the housing when the movable arm has reached the second position and the needle has been loaded into the suturing device **10**. As will be appreciated by those of skill in the art, the engagement of the locking surface of the movable component and the retention surface of the housing (or vice-versa) can prevent the movable arm **120** from returning to the first position to ensure delivery of the needle **135** to the suturing device **10**.

(30) In accordance with a further aspect of the present disclosure, optionally, movement of the movable arm to the second position can activate an indicator confirming that the movable arm has reached the second position. This can indicate to a user that a needle loader **100** has been used and does not contain a sterile needle available for loading into a suturing device **10**. For example, with reference to FIG. **1G**, an indicator **180** is provided that is formed by the cooperative position of a window **182** and a portion of the movable assembly (e.g., **120**) bearing visual indicia **184**. For example, the boom **126** can include indicia formed on the side thereof, such as indicia of two different colors (e.g., green/red), or patterns (e.g., stripes/dots), or characters (e.g., $\frac{1}{2}$, A/B) and the like. One of the two indicia may be displayed through the window prior to the needle **135** being loaded onto the suturing device **10** to signify that the needle loader **100** contains a sterile needle and is ready for use (e.g., via display of a “green” color or the like). The other indicia (e.g., display of a “red” color or the like) may only become visible after the needle is in position to be loaded into the suturing device, or after the needle has actually been loaded into the suturing device. Thus, the visual indicia **180** can confirm that the loading is complete.

(31) It will be appreciated that indicia can be provided in different locations and by way of using

different mechanisms. For example, a different portion of the movable assembly including arm **120** can include indicators in different locations that can be viewable through windows similar to window **184** in different locations. For example, visual indicia can be provided additionally or alternatively on spring **160** that can be visible through a designated window in the housing **110**. If desired, other portions of boom **126**, needle mount **130**, or hub **121** can be provided with such indicia. By way of further example, a portion of the movable arm assembly **120** can cause a visual or tactile indicator formed into the housing **110** to move from a first position to a second position. For example, spring **160** or another movable component coupled to hub **121** can be configured to move a second element, such as a movable stop (e.g., if boss **115** were movable from a first position to a second position) from a first position to a second position. Such movement can include moving a movable button from a first position wherein the button is located inside the housing **110** to a second position wherein the button extends from the housing. Alternatively, the movable assembly can be provided with indicia in multiple locations to provide a plurality of indicators so that at least one of the indicia is visible when viewing the needle loader from multiple angles.

(32) Regardless, after needle loading is complete, the suturing device **10** can then be removed with the needle **135**, and the visual indicia can confirm to personnel that the needle loader has already been used. The needle loader **100** can then be discarded (if the device is a single-use device) or the needle loader **100** can be sent for reprocessing and loading of a new suture needle (if the device is a multi-use device). By way of further example, the indicator **180** can confirm that the movable arm **120** has reached the second position by emitting a sound, a tactile signal, and/or by displaying visual indicia as described. Likewise, the snapping of a lock over a pawl as previously described can also provide an aural indication that the needle **135** has been loaded into the suturing device. The indicator **180** can include a portion that confirms by tactile feel that the movable arm **120** has reached the second position by moving from a first indicator position to a second indicator position such as by protruding through the housing after the needle is loaded. It will be appreciated by those of skill in the art that, if desired, the other embodiments described herein (**200, 300, 400, 500, 600, 700, 800**) may also include visual and/or audible or tactile indicia indicating that a needle has been loaded into an inserted suturing device and/or indication of a spent/used needle loader.

(33) It will be appreciated that the suturing device **10** can be provided in a variety of configurations. Optionally, in some implementations, the suturing device can be provided with a door, hatch or removable cover that can be displaced or removed to expose a needle track of the suturing device, and/or a location in which an arced suturing needle can be deposited by a needle loader. In such cases, the door, hatch or removable cover may be displaced to permit the needle loader to deposit the needle on or in the suturing device.

(34) In one example, the suturing device can optionally include a retractable cover that covers the needle track when in a closed position. As a consequence of engaging the needle loader with the suturing device, the retractable cover may be retracted to facilitate loading of the needle into the track. For example, in accordance with additional aspects of the present disclosure, the needle loader can further comprise a protrusion that extends into a path of a suturing device to be loaded with a needle from the needle loader, wherein the protrusion causes an arced needle track of the suturing device to be exposed as the suturing device is advanced along the path. It will be appreciated by those of skill in the art that the needle loaders described herein can be provided with one or more structural features (bosses, arms, and the like) that facilitate exposing a target location in a suturing device in which to deposit a needle, such as a needle track.

(35) However, it will be appreciated that such a feature is optional. The needle loaders described herein can be configured to engage a suturing device that does not include a retractable cover or door or the like. For example, a suturing device can be provided with a receiving portion for receiving a needle that is configured to deflect or deform to permit the needle to be inserted. For example, the needle may be received by a resilient material disposed on a portion of a suturing

device. Alternatively, the needle (e.g., **135**) can include ferromagnetic material, and the suturing device may include a magnetized portion that attracts and retains the needle in the suturing device with or without the use of a cover to keep the needle from falling out of the device. In another example, a suturing device may have a manually removable cover that is configured to be manually removed by a user or tool before loading the suturing device with a needle.

(36) For purposes of illustration, and not limitation, with reference to FIGS. **1C** and **1D**, the housing **110** of the needle loader **100** can further define an opening **108** therein for receiving the suturing device **10**, wherein the opening **108** at least in part defines the path to insert and load the needle **135** into the suturing device **10**. In some implementations, such as the depicted embodiment, the suturing device **10** can include a retractable cover (illustrated as a sliding cover) to cover the needle track, and the protrusion **152** can retract the retractable cover of the suturing device **10** to expose the arced needle track. The sliding cover of the suturing device **10** is preferably spring loaded so that the cover slides back into place after being withdrawn from the needle loader **100**. In some implementations discussed in further detail below, the protrusion can be coupled to the needle mount instead of the housing **110**. In other implementations, the protrusion can be coupled to the housing (e.g., **110**).

(37) FIGS. **2A** and **2B** depict isometric exterior views of a representative embodiment of a needle loader **200** in accordance with the present disclosure. FIG. **2C** illustrates a view of needle loader **200** that removes a portion of the external housing to view interior components of the loader **200**.

(38) It will be apparent that loader **200** shares a number of features with loader **100** described above. For example, similar to needle loader **100** described above, insertion of the suturing device **10** into needle loader **200** causes pivoting motion of a movable assembly of the needle loader **200** to cause loading of suturing needle **235** into the suturing device **10**. For example, an arm **220** pivots about an axle **226** between a first position wherein a needle mount **230** is spaced apart from a needle track of a suturing device **10** to a second position wherein the needle mount is adjacent the needle track of the suturing device **10** for needle loading. The needle loader **200** includes a housing **210** that defines an elongate passage **208** to receive the suturing device **10**. The elongate passage **208** begins at a first external surface **210a** of the needle loader **200** and is directed inwardly toward the needle mount **230** of the needle loader.

(39) With continuing reference to **2A-2C** and **3A-3D**, in operation, insertion of the suturing device **10** into the needle loader **200** causes the arm assembly **220** including arm **222** with a needle mounted thereon to pivot from a first position where the needle is spaced from the suturing device to a second position that causes the needle to be loaded into the suturing device. As the needle is being loaded into the device, a locking feature **242** engages a catch **212** of the housing **210** of the suturing device **200** to hold the needle mount in place. The suturing instrument (e.g., **10**) can then be withdrawn from the needle loader **200** with the needle loaded therein.

(40) More particularly, and with reference to FIGS. **3A-3D**, the arm **220** includes an upper arm portion **222** that extends away from hub **221** and terminates in needle mount **230**. Needle mount **230** defines thereon a plurality of needle engagement portions **232**, which can include one or more cleats, clips, or the like, to receive a suturing needle therein. Each cleat **232** is mounted on a surface of the mount **230** that faces a needle track **12** of a suturing device **10** as the suturing device **10** approaches the needle **235** as depicted in FIG. **4E**. With reference to FIG. **3A**, needle arm **220** further includes a second arm **224** that extends outwardly and downwardly from a middle region of arm **222**. Arm **224** then executes a bend and traverses a path that is generally parallel to arm **222**.

(41) A lock, or locking feature, **242**, includes a resilient cantilevered arm that extends in a direction opposite from a cam feature **228**, and that terminates in a downwardly extending barb **242a**. Needle loader **200** does not include a discrete spring member (e.g., **160**) but could be modified to include such a spring member to resist forward movement of the suturing device **10** as the suturing device **10** is being loaded. The arm of locking feature **242** deflects upwardly about the point where it is mounted to arm assembly **220** as it passes over catch **212** of housing **210**, and this deflection

creates some resistance as the suturing device **10** engages the needle **235** attached to mount **230**.

(42) Arm **224** terminates in a cam feature **228** at the edge of a platform that traverses a path that is generally parallel to and beneath the needle mount **230**. As depicted in FIG. 3A, the upper surface **228b** of section **228** that faces the needle bearing surface **230a** of the needle mount is actually angled toward that surface to some extent. With reference to FIG. 2C, the barb **242a** of the locking feature **242** slides over a boss, or catch (or another barb) **212** defined in housing **210** as the arm **220** is pushed backwardly. This backward movement is caused by a distal end or end portion of the suturing device **10** urging against a curved or cam surface **228a**. As arm **220** is pushed back by suturing device being advanced along passage **208** of needle loader **200**, the spacing, or gap, between surfaces **228b** and **230a** expands, permitting a distal end portion of the suturing device **10** including needle track **12** to enter the spacing or gap between surfaces **228b** and **230a**. This permits the needle **235** mounted in cleats **232** to align with the needle track **12** of suturing device **10**.

Surface **230a** then is able to align and contact, if desired, an upper surface of suturing device **10** that defines needle track **12** therein. Once so aligned, opposing first and second end sections of needle **235** align with and are received by the needle track **12**, while the cleats **232** continue to hold the needle **235** in place with respect to surface **230a** by way of an interference fit. However, once the first and second end portions of the arcuate needle are received by first and second ends of the arcuate needle track **12**, this provides a sufficient interface to grip the needle so that the suturing device can be withdrawn and release needle **235** from the needle engagement features **232**, illustrated here as fingers, but may include one or more cleats or clips.

(43) The locking feature **242**, in cooperation with catch **212** prevents the needle loader **200** from moving from the second position, wherein the needle is deposited in the suturing device **10**, back to the first position. As such, the needle arm **220** remains in the second position and does not “follow” the suturing device **10** as the suturing device **10** is withdrawn from needle loader **200** because, at this point, barb **242a** has engaged catch **212**, locking arm **220** in place. To accommodate the movement of locking against catch **212**, the arm of lock **242** deflects upwardly to permit the barb **242a** to clear the catch, or cleat **212**.

(44) FIG. 2C presents an isometric view showing the relative positioning of components as suturing device **10** is advanced along passageway **208** prior to contacting cam surface **228a** of arm **224**. FIG. 4D shows an overall view from a lower perspective showing device **10** after suturing device **10** has contacted surface **228a**, and FIG. 4E shows an enlarged view of a portion of FIG. 4D illustrating relative positioning of the needle **235** with respect to the needle track **12** of the suturing device.

(45) FIGS. 4A and 4B show isometric views of the inner surfaces of the two halves of the housing **210**. Each housing half is characterized by an inwardly facing wall including components (tabs, ridges, and the like) extending therefrom to support and guide operation of components of the needle loader **200**. Each housing half defines a socket **216** to receive a respective end of axle **226** of arm **220**. Each side of the housing includes an arcuate ridged guide **214** that extends inwardly from each side of the housing to support the movement of arm **220** as it pivots and prevents the arm **220** from buckling out of its intended plane of travel. One half of the housing **210** includes first and second tabs **214a**, **214b** to act as stops for the arm **220** to limit its range of travel. Window **284** is provided in one of the housing portions to permit a user to view engagement of the needle **235** with the needle track **12** of the suturing device **10**.

(46) For purposes of illustration, and not limitation, with reference to FIGS. 4A-4C, the housing **210** of the needle loader **200** can further define an opening leading to a passageway **208** thereinto to receive the suturing device **10**. In the depicted embodiment of a needle loader **200**, optionally, the suturing device **10** can include a retractable cover (illustrated as a sliding cover) to cover the needle track, and a tip **252** of a protrusion **250** coupled to an inner portion of the housing **210** can cause the retractable cover of the suturing device **10** to be retracted to expose the arced needle track for needle loading. More particularly, and with continuing reference to FIG. 4C, protrusion **250** is

provided that extends in a cantilevered fashion extending downwardly from the housing from a fixed end **254** and at an angle toward the opening that leads to passageway **208** so as to intercept the suturing device **10** as the suturing device proceeds into the needle loader **200** along the path defined by the passage **208**. A free end including a tip **252** of the protrusion **250** is located to contact an optionally slidably displaceable cover of the suturing device **10** to force the cover of the suturing device to retract backwardly as the suturing device **10** is advanced into passage. Protrusion **250** is canted against the direction of travel of the suturing device **10**, and may flex about location **254**, which can function as a hinge and deflect backwardly to some extent without being pushed out of the way and bent fully backward by advancement of the suturing device **10**. Needle loader **200** can be provided with any desired features as described elsewhere herein, including but not limited to an indicator to indicate that the needle has been loaded, and/or that the needle loader has been used, as with the embodiment of FIG. 1.

(47) An alternative embodiment of an arm assembly **220'** is depicted in FIGS. 3E-3H that can be used in combination with a suitably configured housing of a needle loader similar to housing **210** of needle loader **200**. It will be appreciated that corresponding components of housing **210** would need to be relocated or reconfigured somewhat to receive arm assembly **220'**, and that arm assembly **220'** is not configured to be a drop-in replacement for arm assembly **220**. In operation, arm assembly **220'** functions similarly in a number of respects to arm assembly **220**. Insertion of the suturing device **10** into a needle loader including arm assembly **220'** including arm **222'** with a needle mounted thereon causes arm assembly **220'** to pivot from a first position where the needle is spaced from the suturing device to a second position that causes the needle to be loaded into the suturing device. As the needle is being loaded into the device **10**, a locking feature **242'** engages a catch (similar to catch **212** of the housing **210**) to hold the needle mount **230'** in place against a distal end portion of the suturing device **10**. The suturing device (e.g., **10**) can then be withdrawn from the needle loader **200** with the needle loaded therein.

(48) More particularly, and with continuing reference to FIGS. 3E-3H, the arm **220'** includes an upper arm portion **222'** that extends away from a hub **221'** and terminates in needle mount **230'**. Needle mount **230'** defines thereon a plurality of needle engagement portions **232'**, which can include one or more cleats, clips, or the like, to receive a suturing needle therein. Each needle engagement portion **232'** is mounted on a surface of the mount **230'** that faces a needle track **12** of a suturing device **10** as the suturing device **10** approaches the needle. Arm assembly **220'** further includes a second arm **224'** that extends outwardly and downwardly from a hub **221'**. Hub **221'** serves as a mounting point for arms **222'** and **224'** and defines a bearing surface or bore hole **226'** to be received by a corresponding axle or boss of a housing of a needle loader (not shown).

(49) A lock, or locking feature, **242'**, includes a resilient cantilevered arm that extends in a direction that is the same as from a cam feature **228a'**, and that terminates in a downwardly extending tip. Needle arm assembly **220'** does not include a discrete spring member (e.g., **160**) but can be modified to include such a spring member to resist forward movement of the suturing device **10** as the suturing device **10** is being loaded. The cantilevered arm, or paddle, of locking feature **242'** deflects upwardly about the point where it is mounted to lower arm **224'** so that, as the free end of feature **242'** passes over a bump or a catch of a matching housing, the feature **242'** passes over the bump or catch and then snaps in place behind it as the suturing device **10** engages a needle attached to mount **230'**.

(50) Arm **224'** terminates in a pair of cam features **228a'** that traverses a path that is generally parallel to and beneath the needle mount **230'**. The cam features interact with (and are pushed by) a distal face of the suturing device **10** as the device is advanced into the needle loader, causing the arm assembly **220'** to move downwardly until the needle is deposited in the suturing device **10**. As the locking feature **242'** slides over a respective boss or catch, a needle mounted in clips or cleats **232'** aligns with the needle track **12** of suturing device **10**. Once so aligned, opposing first and second end sections of the needle align with and are received by the needle track, while the cleats

232' continue to hold the needle in place. However, once the first and second end portions of the arcuate needle are received by first and second ends of the arcuate needle track, this provides a sufficient interface to grip the needle so that the suturing device can be withdrawn and release needle from the needle engagement features **232**.

(51) The locking feature **242'**, in cooperation with a corresponding catch in a housing (not shown) prevent the arm assembly **220'** from moving from the second position, wherein the needle is deposited in the suturing device, back to the first position. As such, the needle arm assembly **220'** remains in the second position and does not “follow” the suturing device as the suturing device is withdrawn from needle loader because, at this point, locking feature **242'** has engaged a corresponding catch, locking arm assembly **220'** in place.

(52) FIGS. 5A-5O depict aspects of an embodiment of a needle loader **300** in accordance with the present disclosure. Embodiment **300** includes various similarities to needle loader **200**. As depicted, housing **310** is divided into two halves that can each be individually molded, and include various structures to facilitate operation of the device **300**, such as guide, bosses, sockets, and the like. The movable arm **320** of the needle loader **320** is depicted as being rotatable about a plurality of pivot points **321a**, **321b** between a first position wherein a needle mount **330** including a needle **335** is displaced from a needle track **12** of a suturing device, and a second position (e.g., FIG. 5C). Arm **320**, as depicted comprises a four-bar linkage, wherein two bars of the linkage are formed by each of arms **322** and **324**, a third bar is formed by the portion of housing **310** that connects the two sockets that rotatably receive first, upper ends of arms **322**, **324**, and the fourth bar is comprised of needle mount **330**, which is hingedly coupled to arms **322**, **324** at hinge points **327a**, **327b**. The four-bar linkage of the movable arm **320** is rotatable about two pivot points coincident with axles **321a**, **321b** that are received in corresponding sockets in housing **310**, and as a consequence, rotation about those pivot points permits the arms **322**, **324** to remain largely parallel when the arm **320** is articulated, and also helps keep the needle bearing surface of mount **330** to be parallel to the needle track **12** of the suturing device **10**.

(53) In operation, needle loader **300** receives a suturing device by way of an elongate passage **308** that initiates at an outer (preferably circular) opening **308a** defined in an outer surface of the housing **310** of the needle loader **300**. The suturing device **10** is advanced along passage **308** until a free end **352** of a protrusion extending from an inner wall of the housing into the passage **308** contacts concave outer surface of a retractable cover **14** of suturing device **10**. This causes the cover **14** to be held in place while the other portions of the suturing device **10** continue to advance toward the needle **335** disposed on the needle mount **330**. As a curved or ramped distal end face surface **328a** of lower platform **328** contacts a distal region of the suturing device **10**, the orientation of surface **328a** causes arm **320** to be pushed backward and downward. Surface **328a** remains in contact with a concave end face of the suturing device as loading is occurring. At this point, the downwardly facing surface **330a** of the needle mount, which can hold the needle in place with cleats or other suitable means, comes into close proximity with the suturing device, and the needle **335** is received in the needle track **12** of the suturing device **10**.

(54) FIGS. 5A and 5B present cross-sectional views of this procedure, wherein the cross section is viewed down the longitudinal center of the suturing device **10** to illustrate how end face **328a** (see also FIG. 5J) is received into an arcuate gap defined at the end of the suturing device **10**, and how surface **328a** contacts the concave surface at the apex of the arcuate gap of the suturing device **10**. As is visible in the drawings, the needle mount **330** comprises a first surface **330a** defining a boss or retainer thereon to receive the arced needle **335**. Surface **328b** faces the first surface **330a** of the needle mount. The surface **330a** and surface **328b** therefore cooperate to define a gap therebetween to receive a distal portion of the suturing device **10**. Surface **328b** comprises a ramped surface to guide the arced needle track **12** of the suturing device **10** toward the arced needle **335** mounted on the needle mount **330** as the suturing device **10** is advanced along a path of travel with respect to the needle loader **300**. After receiving the needle **335**, the suturing device **10** can be withdrawn

from the needle loader **300**.

(55) FIGS. 5D-5I provide additional views of the arm assembly **320** from a variety of views. FIGS. 5D-5I present a right-side elevational view, a back elevational view, a left-side elevational view, a front elevational view, a top plan view, and a bottom plan view, respectively. FIGS. 5J-5O present a front left upper isometric view, a bottom left isometric view, a top right isometric view, a bottom isometric view, a lower front isometric view, and a lower back isometric view, respectively.

Corresponding components of arm assembly **320** are labeled with identical reference numbers.

(56) With continuing reference to FIGS. 5D-5O, arm assembly **320** includes a first axle **321a** coupled to upper arm (relatively upper per FIG. 5A) and a lower axle **322b** coupled to lower arm **324**. As is shown in FIGS. 5J and 5M, surface **328a** includes a convex surface to engage a distally facing surface within the concavity of suturing device **10**. Locking features **342a**, **342b** and **342c** are also visible. Locking features **342a** and **342b** are locking features in the form of deflectable paddles, wherein each paddle extends from an arm that wraps around element **328**. Specifically, a bent arm, or elbow extends laterally out of each side of element **328** that wraps around and enlarges. During operation of needle loader **300**, the free ends of the locking features **342a**, **342b** slide past a catch, cleat or boss (not shown) inside of housing **310**. If desired, the locking features **342a**, **342b** can be captured by a window defined in either wall of the housing (not shown) to verify when the needle and needle mount have reached a second position wherein the needle is able to engage the suturing device **10**. Each outwardly facing surface of each paddle **342a**, **342b** can additionally be provided with visual indicia such that the indicia could be caused to overlap with an opening in the housing, as described above with respect to embodiment **100**, to indicate if the needle loader **300** has been used, or whether it has already been used, and can be processed further to reload or dispose of it. It will be further appreciated that paddle **342a**, **342b** may simply be used as stabilizers to slide against inner walls of the housing **310** and not act as a locking mechanism. With continuing reference to FIGS. 5D-5O, arm assembly **320** can further include an additional locking feature **342c** that can be configured to pass over a boss or a catch or a cleat after the arm assembly **320** has moved from the first position to the second position.

(57) Needle loader **300** can be provided with any desired features as described elsewhere herein, including but not limited to a lock to lock the arm **320** in place after the needle has been loaded onto the suturing device, and an indicator to indicate that the needle has been loaded, and/or that the needle loader has been used, as with the embodiment of FIG. 1.

(58) In further accordance with the disclosure, needle loader devices can be provided that utilize a sliding action to translate an arm assembly bearing a suturing needle from a first position to a second position. For example, insertion of the suturing device can cause a movable assembly or arm assembly of the needle loader to be moved relative to the housing along guide tracks defined in the needle loader housing to effectuate needle loading. This can be accomplished in a variety of ways.

(59) For example, as illustrated in FIGS. 6A-6E a further embodiment of a needle loader **400** is presented in accordance with the present disclosure, or aspects thereof. Embodiment differs from the previous embodiments (**100**, **200**, **300**) in several respects. For example, needle loader includes a slidable arm, or arm assembly, **420** having bosses or guides **421** that slide in and are guided by one or more tracks **411** (FIG. 6D) defined by the housing **410**. While the tracks **411** are depicted as being linear to provide linear motion, the track **411** can also be curved or arced, as desired. The arm assembly **420** moves from a first position (FIG. 6A) wherein the arm assembly **420** is closer to the opening **408a** that partially defines the passageway **408**, to a second position (FIGS. 6B, 6C) wherein the arm **420** has slid along the tracks **411** on guides **421** and the needle has been loaded into the suturing device **10**.

(60) With reference to FIG. 6E, arm or arm assembly **420** includes a main hub **423** that includes a plurality of bosses defined on each side thereof to be guided by corresponding tracks defined in the housing **410**. Hub **423** includes a plurality of arms extending therefrom. An arm **422** extends from a

lower portion of the hub **423** that terminates in a needle mount **430** having one or more needle retention features, such as fingers, clips or cleats **432** to hold a suturing needle (not shown) in place. A suture retention feature, clip, finger or cleat **447** can be provided to hold the suture that is attached to the suturing needle to help hold the suture out of the way as the needle loader **400** is loading a suturing device with a suturing needle. A second arm **424** extends outwardly from the hub **423** and extends over and around arm **422** and the needle mount **430**. Arm **424** then extends downwardly and terminates in a curved surface **428a** that rides over the top of the distal end of the suturing device **10** when it is inserted, and downwardly extending protrusion **452** acts to engage the cover of the suturing device to cause the cover to be pushed back to expose the needle track. Arm **442** also extends outwardly from the hub **423** and terminates in a locking feature **442a**, which may be in the form of a barb to engage with cleat or boss **412** defined in the housing **410** (FIGS. **6C**, **6D**).

(61) A fourth arm, or spring, **460** also extends outwardly from the hub, and terminates in axle **461** that is received within sockets **416** defined in the housing. FIG. **6C** shows the barb **442a** engaged with the cleat **412**, but shows axle and spring **460** displaced with hub **423** for purposes of illustration. In actual operation of the needle loader **400**, the axle **461** remains in sockets **416**, so that as the hub **423** and arms are pushed further into the housing by suturing device **10** along tracks **411**, spring **460** is stretched and deformed to provide a countervailing force against the advancement of the suturing device **10**. As further shown in FIG. **6E**, arm **424** can be provided with a guide or boss **421** to ride in a dedicated track **411** defined in the inner surface of the housing **410**. Tracks **411** are illustrated in FIG. **6D**. FIG. **6B** shows the arm assembly **420** at its starting point. FIG. **6A** shows the relative position of the arm assembly **420** and the suturing device **10** after the needle has been delivered to the suturing device **10**, but illustrates the spring axle **461** as not being engaged with the socket **416**.

(62) In use, needle loader **400** is used by inserting a distal end of the suturing device **10** into the opening **410** until surface **428a** contacts the sliding cover of the suturing instrument **10**. It should be appreciated that the sliding cover is optional and may not be present in some embodiments. The further advancement of suturing device **10** along passage **408** pushes arm assembly **420** further into the housing (to the left as illustrated in the drawings) against the force of spring **460**, which is being deformed and straightened in the process, thereby acting as a tension spring, and the slanted directionality of the tracks **411** cause the arm assembly **420** to move downward as well as along the direction of travel of the suturing device **10**. As the needle mount **430** travels in this matter, it approaches the needle track along a vertical direction until the needle is pushed into the arced needle track of the suturing device. By this time, barb **442a** has slid along and engaged boss **412**, preventing backward movement of the arm assembly **420** as the suturing device is withdrawn.

(63) Needle loader **400** can be provided with any desired features as described elsewhere herein, including but not limited to an indicator to indicate that the needle has been loaded, and/or that the needle loader has been used, as with the embodiment of FIG. **1**.

(64) In further accordance with the disclosure, further embodiments of a needle loader are provided wherein insertion of the suturing device into the needle loader causes sliding movement of an arm assembly that includes a suturing needle to be loaded into the suturing device from a first position to a second position. Optionally, such devices can be configured to facilitate removal or retraction of a cover of a needle track of the suturing device. Also, optionally, the needle loader can include a predetermined length of suturing material of various types (conventional, barbed, absorbable, and the like) that may be held in place in a suture compartment and/or wound around one or more bosses. The other embodiments herein can be provided with suturing material disposed in a suture compartment and/or wound around a predetermined path within a portion of the needle loader.

(65) For purposes of illustration only, and not for purposes of limitation, FIGS. **7A-7N** illustrate another embodiment of a needle loader **500** in accordance with the present disclosure, or aspects thereof. Needle loader **500** is generally similar in a number of aspects to the previous embodiments.

In particular, the general configuration of loader **500** greatly resembles loader **400**, in that loader **500** includes an arm assembly **520** with guides or bosses **521** that ride in corresponding tracks **511** formed in the housing **510** of the loader **500**.

(66) However, needle loader **500** additionally includes a suture compartment **590** to hold a length of suture that is coupled to the arced needle that is disposed in the loader **500**. The length of suture can be, for example, between about six and about 24 inches in length. As illustrated, the suture compartment **590** is coupled to the hub **523** of the arm assembly **520**. In the illustrated embodiment, the suture compartment is integrally formed with the arm assembly **520**, but compartment may instead be formed into the housing **510**, or may be a separate component. The suture compartment **590** can be coupled to the needle mount **530**, if desired. If desired, the suture compartment **590** can be integrally formed with the needle mount. As depicted, the suture compartment **590** can be oblong in shape. In some implementations, the suture compartment can be defined by a peripheral curved wall. The peripheral curved wall can be circular in shape, elliptical in shape, or a combination of different shapes. An example of such an elliptical or circular suture compartment **690** can be seen with reference to needle loader **600** depicted in FIGS. **8A-8C**, discussed in further detail below.

(67) FIG. **7G** illustrates relative placement of components in the needle loader **500** prior to insertion of the suturing device **10** therein. When the instrument **10** is inserted into passageway **508**, surface **528a** urges against the suturing device **10** and the arm assembly **520** is pushed further into the housing **510**, which also causes spring **560** to start compressing and urges against the suturing device **10** with a force that increases as the suturing device is inserted further into passageway **508**. The protrusion **552** (FIG. **7E**) then engages the optional cover of the device **10**, causing the cover to be retracted due to the force delivered by the spring **560**. As the suturing device is pushed still further in to passageway, upper surface **529a** begins to ride along boss **517**, and the upwardly inclined angle of surface **529** then causes arm **524** and needle mount **530** to be pushed downward toward the needle track of the suturing device, resulting in alignment of the needle with the needle track, and then deposition of the needle into the needle track of the suturing device **10**. Relative placement of the suturing device and the loader **500** can be seen in FIGS. **7H-7J**. A locking feature **542a**, which may be in the form of a barb (see FIG. **7B**), then slides over a corresponding cleat **512** formed into the housing **510**. Engagement of this locking mechanism thereby prevents the arm assembly **520** from moving with the suturing device **10** as the suturing device is withdrawn. Similar to embodiment **400**, spring **560** includes an axle **561** that is received by one or more sockets **516** defined in the housing. However, spring **560** functions as a compression spring, rather than a tension spring.

(68) FIG. **7A** illustrates relative lateral placement of the arm assembly **520**, but does not illustrate deflection of arm **524** with respect to boss **517** for purposes of illustration. As illustrated, FIG. **7B** shows the relative location of hub **523** to boss or guide **521**, and illustrates openings **592** in suture compartment to receive a suture winding fixture **698** as illustrated in FIGS. **7K-7N** to dispose suture **699** in the suture compartment **590**. The suture compartment **590** of FIG. **7K** is illustrated showing a hinged cover **595** attached thereto that can be closed upon suture being deposited therein that is attached to a needle **535**. The cover **595** can also be a separate component. Winding bosses **698a** are defined on winding fixture **698**, which are slid through openings **592** in suture compartment **690**. Suture **699** is then wrapped around the bosses **698a** to properly align suture in the compartment **590**. The fixture **698** can be configured to snap over the suture compartment or simply align with it. Once the suture is wrapped, the fixture can be removed, and the cover **595** can be closed, and the loader **500** can be assembled, packaged and sterilized.

(69) FIGS. **8A-8C** present a needle loader **600** that is a variation of loader **500**. For example, loader **500** provided an oblong suture compartment **590**, whereas embodiment **600** provides a modified design that is generally circular in shape. If desired, the compartment can be elliptically shaped. Needle loader **600** can include one or more contoured gripping surfaces **603** formed into the

housing **610** to facilitate gripping needle loader **600** by a user when installing a needle on suturing device **10**.

(70) In addition, needle loader **600** illustrates an optional visual indicator **680** at a top edge thereof, wherein a first indicator **682a** having indicia reflecting that the loader **600** has not been used, is aligned with a window through the housing **610** prior to using the loader. When the suturing device has been inserted into opening **608** and the loader is used, the assembly including the arm **624** and the suture compartment **690** is slid to the left (with respect to FIG. **8A**), and a second indicia **682b** is visible through the window, indicating that the loader **600** has been used. Indicia **682b** is formed onto a deflectable paddle that passes over a cleat **612** formed into the housing (FIG. **8C**), and holds the inner assembly of the loader **600** in place. A spring **660** is also provided in the form of a compressive spring that can have any one of a variety of shapes. For example, the spring **660** need not rigidly anchor to the housing **610** because it is being compressed. Thus, while a socket **616** can be provided to receive a free end of the spring **660**, the spring can be configured as illustrated in FIG. **8B** and urge against the inner surface of the housing **610** of the loader **600**. The arm **624** is deflected by sliding over boss **617** to help advance the needle mount **630** downwardly toward the needle track **12** of the suturing device **10** in a manner similar to embodiment **500**. Suture compartment **690** includes guides or bosses **621** thereon to ride in tracks **611** defined on an interior surface of the housing **610**.

(71) The disclosure provides still further implementations of a needle loader in accordance with the present disclosure, such as the embodiment illustrated in FIGS. **9A-9P**. In such implementations, the needle loader can be a low-profile compact needle loader **700** to facilitate stacking in a container with other needle loaders to optimize space for shipment. Such needle loaders can provide a discrete compartment **710c** that includes a predetermined length of suture that is attached to the arced suturing needle. In use, such a low-profile loader **700** can be aligned with a distal end portion of the suturing device (e.g., **10**) and the suturing device **10** can be aligned with and introduced into a passageway **708** of the suturing device. A movable arm assembly, or movable inner assembly **720** can receive the suturing device **10** and act to displace an optional displaceable cover of the suturing device as the suturing device is advanced into the needle loader. As the suturing device is further advanced into the passageway, an arm **722** carrying a needle mount **730** with a needle removably coupled thereto can be moved into contact with a needle track of the suturing device due to an inclined surface **722a** of the arm **722** sliding along a stationary boss **717** that causes the arm **722** to bend toward the needle track of the suturing device. As the needle is placed in the suturing device **10**, optionally, a lock or locking element **742** can be engaged by an engagement element **712**, such as a cleat, edge, boss, opening in the housing, or the like. The movable assembly **720** may be held in place by the engagement of the locking element **742** and the engagement element **712**. With the suturing needle engaged by the needle track of the suturing device **10**, the suturing device **10** may be withdrawn from the needle loader **700** and the arced suturing needle disengages from the needle mount **730**. Suture material (not shown) can be removed from the suture compartment **710c** through a groove **709** that is defined along the face of the needle loader proximate the opening **708** that connects to the suture compartment **710c**. However, the housing overall has considerable stiffness to prevent groove **709** from closing if needle loader **700** is squeezed laterally from both sides. Suturing material can then be unwound from the needle loader **700** either before or during the surgical procedure. The opening **708** can be provided with one or more ridges that extend along the longitudinal axis of opening **708** to accommodate mating grooves in the suturing device that cooperatively function as an alignment feature. By way of further example, the alignment feature can be an upper groove **707** that passes into the bottom surface of the needle loader to accommodate an alignment feature of the suturing device **10**, such as a keel, or lower ridge, of the suturing device. Any of these alignment features can help facilitate rotational alignment with the suturing device **10** and the needle loader **700** about a longitudinal axis of the suturing device **10** to help ensure that the arced suturing needle

successfully mates with the suturing device **10**.

(72) With continuing reference to the Figures, FIGS. **9A-9B** show an upper left side and lower right front side perspective views, respectively, of needle loader **700** in accordance with the present disclosure. For purposes of embodiment **700**, the flattened surface illustrated in FIG. **9A** is referred to herein as the “top” surface for purposes of reference. FIGS. **9C** and **9D** show the assembly in an inverted view to facilitate illustration, wherein FIG. **9C** illustrates a sectioned view down a longitudinal direction of the loader **700** located to the “right” side of the longitudinal axis, wherein the needle loader **700** is upside down. Needle loader **700** can also include one or more contoured gripping surfaces **703** formed into the housing **810** to facilitate gripping needle loader **700** by a user when installing a needle on suturing device **10**.

(73) The housing **710** of the needle loader is composed of three main components. First, an upper concave shell portion **710e** cooperates with an intermediate plate section **710d** to define a cavity to receive a longitudinally movable assembly, or movable arm assembly, **720**. The assembly **720** is configured to slide backward into the cavity defined by portions **710d**, **710e** when pushed by a suturing device **10**. Forward motion of the suturing device **10** is resisted by a compression spring **760**. Spring **760** traverses a serpentine path similar to springs **560** and **660**.

(74) Housing component **710d** and housing component **710a**, or cover plate, cooperate to form suture compartment **710c**. While the entirety of the housing **710** can be made from transparent material, providing cover plate **710a** as a transparent member can permit a user to view suture that is present inside of the suture compartment **710c**. Component **710a** is a plate shaped member having a pair of suturing winding openings **710b** configured to receive a pair of suture winding bosses of a suture winding fixture (not shown) that is similar in form and function to suturing winding fixture **698**. In use, bosses of the fixture are introduced through the openings **710b**, and suture is wound around the bosses prior to installing plate **710a** into a peripheral seat defined by the peripheral wall of component **710d**. The suture can simply be wound around the bosses in an oval pattern, or can be crossed over to form a figure-8 pattern, as may be desired. After the suture is placed inside compartment **710c**, the suture fixture can be removed from openings **710b**, leaving the coiled suture material in place. The suture material is coupled at one end to the suturing needle, which is removably coupled by needle engagement elements **732** such as fingers, clamps, cleats and the like to suture mount **730**. The suture passes from the needle through groove **709**, which passes through a portion of plate **710d** near the entrance **708** of the needle loader **700** and extends to the face of the device at the entrance of the loader **700**, to permit the needle loader **700** to receive the arced suturing needle and have the suture trailing from the needle pass through the portion of groove **709** that passes through the plate **710d**. When the needle loader **700** has deposited the suturing needle into the suturing device **10**, the suturing device can be withdrawn, and the suture that trails from the needle passes into the suture compartment, and can be withdrawn through the portion of groove **709** that extends from the entrance **708** of the needle loader **700** to the suture compartment **710c**.

(75) With continuing reference to FIG. **9C**, as the suturing device **10** is advanced into opening **708**, assembly **720** is pushed back into the housing **710**, and inclined surface **722a** of arm **722** rides over the upper surface of boss **717**. This causes the needle mount **730** bearing the needle to be pushed into contact with the distal end portion of the suturing device to place first and second ends of the needle into first and second ends of the needle track. When this engagement has completed, arm assembly **720** has been pushed back far enough into the housing **710** to permit the optional locking element **742** to slide over and engagement element **712** and snap in place, holding arm assembly **720** in place with spring **760** in a compressed state.

(76) FIG. **9D** presents a lower perspective view of the arm assembly **720**. Arm assembly includes a central portion **723** to which other components of the arm assembly attach. Central portion **723** includes a backwardly extending spring **760** attached thereto that traverses a serpentine path that resists a longitudinal compressive load. The spring **760** has a cross sectional shape that is crescent

shaped, which is best viewed in the end view of FIG. 9P. This shaping is to permit locking feature 742 to prevent collision with the windings of spring 760 as spring 760 is compressed.

(77) With continuing reference to FIG. 9D, a pair of arms 724 extend in a direction opposite the direction of the spring, toward the opening of the passageway 708 of the needle loader 700. Each arm 724 is generally planar in shape and slides against inner walls of the housing 710. Each arm 724 terminates in a protrusion 752 that receives a distal end portion of the suturing device and urges against the sliding cover of the suturing device 10 to displace the cover and expose the needle track as the suturing device 10 is being advanced along the passage 708. The protrusions 752 at the free end of each arm 724 define a gap 729 therebetween to permit a suture to pass through. Gap 729 cooperates with gap 709 to permit the suture to pass therethrough that is attached to the suturing needle and helps prevent interference as the suturing device 10 is being withdrawn from the needle loader 700 after the suturing device 10 has been loaded with an arced suturing needle. Elongate arm 722 also extends proximally from the central region 723 of the arm assembly 720. Arm 722 terminates in needle mount 730, as described above. Arm 722 includes an inclined surface 722a to facilitate movement of the needle mount 730 from a first position displaced from the suturing device along a direction orthogonal to the direction of travel of the suturing device 10 along the passage 708 to a second position wherein the needle mount 730 contacts the distal end portion of the suturing device 10. Stated another way, provided with the spatial reference frame wherein the needle loader 700 is “upright” in FIG. 9A, during loading, the inclined surface 722a of the arm 722 pushes the needle mount with needle mounted thereon downwardly into contact with the suturing device 10 in the orientation in FIG. 9A (corresponding to upward motion in the orientation depicted in FIG. 9C).

(78) FIGS. 9E-9F present upper isometric views of the arm assembly 920. Locking feature 942 is disposed at the top of the arm assembly, when considering that the device 700 is upright in FIG. 9A, and the spring 760 slides along a bottom of the housing 710 of the needle loader. With reference to FIG. 9G, a proximal end face 724a of arms 724 is shown, each of which terminates in a protrusion surface 752 that engages the cover of the suturing device 10, and a pair of ramped features 754 that urge against the end face of the retractable cover of the suturing device. FIGS. 9H-9J present aspects of a cross sectional view of the movable arm assembly 720 of the suturing device 700. The plane specifying the cross section is specified in FIG. 9H, and upper and lower isometric section views are shown illustrating relative placement of the needle mount 730 and the needle retention features 732. FIGS. 9K, 9L, 9M, 9N, 9O, and 9P present a top plan view, a bottom plan view, a right-side elevational view, a left side elevational view, a front side elevational view and a rear side elevational view, respectively, of arm assembly 720.

(79) The disclosure provides additional implementations of a needle loader 800 in accordance with the present disclosure illustrated in FIGS. 10A-11F. Needle loader 800 presents an implementation of an ergonomic needle loader with a rounded back end to fit in the palm of the hand of a user. Needle loader 800 can also include one or more contoured gripping surfaces 803 formed into the housing 810 to facilitate gripping needle loader 800 by a user when installing a needle on suturing device 10. Needle loader 800 optionally can include a discrete compartment 890 that includes a predetermined length of suture 899 that is attached to the arced suturing needle. In use needle loader 800 is aligned with a distal end portion of the suturing device (e.g., 10) and the suturing device 10 can be aligned with an introduced into a passageway 808 of the suturing device. A movable arm assembly, or movable inner assembly 820 can receive the suturing device 10 and act to displace a displaceable cover of the suturing device as the suturing device is advanced into the needle loader.

(80) As the suturing device 10 is further advanced into the passageway 808, an arm 822 carrying a needle mount 830 with a needle removably coupled thereto can be moved into contact with a needle track of the suturing device 10 due to an inclined surface 822a of the arm 822 sliding along a stationary boss 817 that causes the arm 822 to bend toward the needle track of the suturing

device. As the needle is placed in the suturing device, lock or locking elements **842a**, can be engaged by an engagement element **812** (FIG. **11D**), such as a cleat, edge, boss, opening in the housing, or the like. The movable assembly **820** may be held in place by the engagement of the locking element **842a** and the engagement element **812**. With the suturing needle engaged by the needle track of the suturing device **10**, the suturing device **10** may be withdrawn from the needle loader **800** and the arced suturing needle disengages from the needle mount **830**. Suture material (not shown) can be removed from the suture compartment **890** through a groove **809** that is defined along the face of the needle loader proximate the opening **808** that connects to the suture compartment **890**. Suturing material **899** can then be unwound from the needle loader **800** either before or during the surgical procedure.

(81) With continuing reference to the Figures, FIGS. **10A-10C** show a lower left perspective view, a lower front perspective view and a left side schematic view, respectively, of needle loader **800**. For purposes of reference herein, embodiment **800** is disposed in an “upright” position in FIGS. **10A-10C**.

(82) The housing **810** of the needle loader **800** is composed of two generally symmetrical halves, wherein the left side half is illustrated in FIGS. **10H-10I**. The two housing halves cooperate to form a cavity to receive the movable assembly **820**, and a second compartment **890** to receive the suture **899**. The arm assembly **820** is configured to slide backward into the cavity defined by housing portions when pushed by a suturing device **10**. Forward motion of the suturing device **10** is resisted by a compression spring **860**. Spring **860** traverses a serpentine path similar to springs **560 660**, and **760**.

(83) The suture material **899** is coupled at one end to the suturing needle, which is removably coupled by needle engagement elements **832** such as fingers, clamps, cleats and the like to suture mount **830**. The suture **899** passes from the needle through groove **809**, which passes through a gap defined between the two portions of the housing **810**. This arrangement permits the needle loader **800** to receive the arced suturing needle and have the suture trailing from the needle pass through the groove **809** after the suturing device **10** is withdrawn from the needle loader **800**. Consequently, after the needle loader **800** has deposited the suturing needle into the suturing device **10**, the suturing device **10** can be withdrawn, and the suture **899** that trails from the needle passes into the suture compartment and can be withdrawn through the portion of groove **809**.

(84) With continuing reference to the figures, as the suturing device **10** is advanced into opening **808**, assembly **820** is pushed back into the housing **810**, and inclined surface **822a** of arm **822** rides over the upper surface of boss **817**. This causes the needle mount **830** bearing the needle to be pushed into contact with the distal end portion of the suturing device to place first and second ends of the needle into first and second ends of the needle track of the suturing device **10**. When this engagement has completed, arm assembly **820** has been pushed back far enough into the housing **810** to permit the locking element **842a** to slide over and engagement element **812** and snap in place, holding arm assembly **820** in place with spring **860** in a compressed state. It will be appreciated that a surface of locking element **842a** can act as one or more visual indicium if engagement element **812** is a window formed into the wall of the housing **810**, as illustrated in FIG. **11D**.

(85) Arm assembly **820** includes a central portion **823** to which other components of the arm assembly **820** attach. Central portion **823** includes a backwardly extending spring **860** attached thereto that traverses a serpentine path that resists a longitudinal compressive load.

(86) With continuing reference to FIGS. **10D-10G**, a pair of arms **824** extend in a direction opposite the direction of the spring, toward the opening of the passageway **808** of the needle loader **800**. Each arm **824** is generally planar in shape and cooperate to define a gap **829** therebetween to accommodate passage of suture **899** that trails from the needle and into the suture compartment **890**. Arms **824** slide against inner walls of the housing **810**. A further boss **842c** is provided on an underside of arm assembly to prevent proximal motion of the arm assembly. Each arm **824**

terminates in a protrusion **852** that receives a distal end portion of the suturing device and urges against the sliding cover of the suturing device **10** to displace the cover and expose the needle track as the suturing device **10** is being advanced along the passage **808**. The protrusions **852** at the free end of each arm **824** also cooperate to define gap **829** therebetween to permit the suture **899** to pass therethrough. Elongate arm **822** also extends proximally from the central region **823** of the arm assembly **820**. Arm **822** terminates in needle mount **830**, as described above. Arm **822** includes an inclined surface **822a** to facilitate movement of the needle mount **830** from a first position displaced from the suturing device to a second position wherein the needle mount **830** contacts the distal end portion of the suturing device **10**. Bosses **821** are provided on arm assembly **820** that are slidably received in respective guide tracks **811** in the inner surfaces of housing **810**.

(87) FIG. **11A** illustrates a further view of needle loader **800** in use, with a packet of suture **899** disposed partially within compartment **890**. FIGS. **11B** and **11C** are partial upper and lower isometric views of arm assembly **824** showing routing of a suture **899** through slot **829**. FIG. **11D** is a side view of a representative implementation of needle loader **800** illustrating relative placement of suture material **899** in compartment **890**.

(88) FIGS. **11E-11F** illustrate a further implementation of a needle loader **900** that is a variation of the needle loader **800**, wherein needle loader **900** includes a streamlined ergonomic housing **910** having one or more gripping contours or surfaces **903**. FIG. **11E** illustrates placement of suturing device **10** into the opening **908** of needle loader **900**, and FIG. **11F** is a front side upper isometric view of needle loader **900** illustrating a gripping surface **903** on the housing **910** of the needle loader and groove **909**, similar to groove **809**, that permits suturing material **999** in a lower compartment to be removed from the needle loader **900** as the suturing material **999** is used. Also depicted is an indicia **905** located on an upper surface of the housing **910** to guide placement and direction of the needle loader **900** with respect to the suturing device **10**. The interior components and layout of needle loader can be the same or substantially the same as needle loader **800**.

(89) Each embodiment of needle loader (**100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**) can be provided with additional features as described in other embodiments, such as lock devices, visual or tactile indicators, suture compartments, and the like. Moreover, each such embodiment can be provided in a sterile kit that is ready to be opened upon use in a procedure. The kit may be a single use device or may be reprocessed and reloaded with a new/sterilized needle. Where possible, components are preferably integrally molded to reduce the number of parts needed for each given device. The needle loaders are preferably made from suitably selected injection molded plastic material.

(90) Embodiments described herein may be used, for example, with remotely operated, computer-assisted systems (such, for example, teleoperated surgical systems) such as those described in, for example, U.S. Pat. No. 9,358,074 (filed May 31, 2013) to Schena et al., entitled “Multi-Port Surgical Robotic System Architecture”, U.S. Pat. No. 9,295,524 (filed May 31, 2013) to Schena et al., entitled “Redundant Axis and Degree of Freedom for Hardware-Constrained Remote Center Robotic Manipulator”, and U.S. Pat. No. 8,852,208 (filed Aug. 12, 2010) to Gomez et al., entitled “Surgical System Instrument Mounting”, each of which is hereby incorporated by reference in its entirety. Further, embodiments described herein may be used, for example, with a da Vinci® Surgical System, such as the da Vinci Xi® Surgical System or the da Vinci Xi® Surgical System commercialized by Intuitive Surgical, Inc., of Sunnyvale, California. Although various embodiments described herein are discussed in connection with a manipulating system of a teleoperated surgical system, the present disclosure is not limited to use with a teleoperated surgical system. Various embodiments described herein can optionally be used in conjunction with hand-held, manual instruments.

(91) As discussed above, in accordance with various embodiments, surgical instruments of the present disclosure are configured for use in teleoperated, computer-assisted surgical systems employing robotic technology (sometimes referred to as robotic surgical systems). Referring now to FIG. **12**, an embodiment of a manipulator system **1200** of a computer-assisted surgical system, to

which surgical instruments are configured to be mounted for use, is shown. Such a surgical system may further include a user control system, such as a surgeon console (not shown) for receiving input from a user to control instruments coupled to the manipulator system **1200**, as well as an auxiliary system, such as auxiliary systems associated with the DA VINCI SI® and DA VINCI XI®, Da Vinci SP, and Ion systems noted above.

(92) As shown in the embodiment of FIG. **12**, a manipulator system **1200** includes a base **1220**, a main column **1240**, and a main boom **1260** connected to main column **1240**. Manipulator system **1200** also includes a plurality of manipulator arms **1210**, **1211**, **1212**, **1213**, which are each connected to main boom **1260**. Manipulator arms **1210**, **1211**, **1212**, **1213** each include an instrument mount portion **1222** to which an instrument **1230** may be mounted, which is illustrated as being attached to manipulator arm **1210**.

(93) Instrument mount portion **1222** comprises a drive assembly **1223** and a cannula mount **1224**, with a transmission mechanism **1234** of the instrument **1230** (similar to suturing device **10** described hereinabove) connecting with the drive assembly **1223**, according to an embodiment. Cannula mount **1224** is configured to hold a cannula **1236** through which a shaft **1232** of instrument **1230** may extend to a surgery site during a surgical procedure. Drive assembly **1223** contains a variety of drive and other mechanisms that are controlled to respond to input commands at the surgeon console and transmit forces to the transmission mechanism **1234** to actuate the instrument **1230**. Although the embodiment of FIG. **12** shows an instrument **1230** attached to only manipulator arm **1210** for ease of viewing, an instrument may be attached to any and each of manipulator arms **1210**, **1211**, **1212**, **1213**.

(94) Other configurations of surgical systems, such as surgical systems configured for single-port surgery, are also contemplated. For example, with reference now to FIG. **13**, a portion of an embodiment of a manipulator arm **2140** of a manipulator system with two surgical instruments **2300**, **2310** in an installed position is shown. The surgical instruments **2300**, **2310** can generally correspond to instruments discussed above, such as suturing device **10** described elsewhere herein. A. For example, the embodiments described herein may be used with a DA VINCI SP® Surgical System, commercialized by Intuitive Surgical, Inc. of Sunnyvale, California. The schematic illustration of FIG. **13** depicts only two surgical instruments for simplicity, but more than two surgical instruments may be mounted in an installed position at a manipulator system as those having ordinary skill in the art are familiar with. Each surgical instrument **2300**, **2310** includes a shaft **2320**, **2330** that at a distal end has a moveable end effector or an endoscope, camera, or other sensing device, and may or may not include a wrist mechanism (not shown) to control the movement of the distal end.

(95) In the embodiment of FIG. **13**, the distal end portions of the surgical instruments **2300**, **2310** are received through a single port structure **2380** to be introduced into the patient. As shown, the port structure includes a cannula and an instrument entry guide inserted into the cannula. Individual instruments are inserted into the entry guide to reach a surgical site.

(96) Other configurations of manipulator systems that can be used in conjunction with the present disclosure can use several individual manipulator arms. In addition, individual manipulator arms may include a single instrument or a plurality of instruments. Further, as discussed above, an instrument may be a surgical instrument with an end effector or may be a camera instrument or other sensing instrument utilized during a surgical procedure to provide information, (e.g., visualization, electrophysiological activity, pressure, fluid flow, and/or other sensed data) of a remote surgical site.

(97) Transmission mechanisms **2385**, **2390** are disposed at a proximal end of each shaft **2320**, **2330** and connect through a sterile adaptor **2400**, **2410** with drive assemblies **2420**, **2430**. Drive assemblies **2420**, **2430** contain a variety of internal mechanisms (not shown) that are controlled by a controller (e.g., at a control cart of a surgical system) to respond to input commands at a surgeon side console of a surgical system to transmit forces to the force transmission mechanisms **2385**,

(98) The embodiments described herein are not limited to the embodiments of FIG. 12 and FIG. 13, and various other teleoperated, computer-assisted surgical system configurations may be used with the embodiments described herein. The diameter or diameters of an instrument shaft and end effector are generally selected according to the size of the cannula with which the instrument will be used and depending on the surgical procedures being performed.

(99) This description and the accompanying drawings that illustrate various embodiments should not be taken as limiting. Various mechanical, compositional, structural, electrical, and operational changes may be made without departing from the scope of this description and the invention as claimed, including equivalents. In some instances, well-known structures and techniques have not been shown or described in detail so as not to obscure the disclosure. Like numbers in two or more figures represent the same or similar elements. Furthermore, elements and their associated features that are described in detail with reference to one embodiment may, whenever practical, be included in other embodiments in which they are not specifically shown or described. For example, if an element is described in detail with reference to one embodiment and is not described with reference to another embodiment, the element may nevertheless be claimed as included in the other embodiment.

(100) For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing quantities, percentages, or proportions, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about,” to the extent they are not already so modified. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

(101) It is noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the,” and any singular use of any word, include plural referents unless expressly and unequivocally limited to one referent. As used herein, the term “include” and its grammatical variants are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that can be substituted or added to the listed items.

(102) Further, this description's terminology is not intended to limit the invention. For example, spatially relative terms—such as “beneath”, “below”, “lower”, “above”, “upper”, “proximal”, “distal”, and the like—may be used to describe one element's or feature's relationship to another element or feature as illustrated in the figures. These spatially relative terms are intended to encompass different positions (i.e., locations) and orientations (i.e., rotational placements) of a device in use or operation in addition to the position and orientation shown in the figures. For example, if a device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be “above” or “over” the other elements or features. Thus, the illustrative term “below” can encompass both positions and orientations of above and below. A device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(103) Further modifications and alternative embodiments will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the devices and methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various embodiments shown and described herein are to be taken as illustrative. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and

certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the spirit and scope of the present teachings and following claims.

(104) It is to be understood that the particular examples and embodiments set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings.

(105) Other embodiments in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as illustrative and for example only, with the following claims being entitled to their fullest breadth, including equivalents, under the applicable law.

Claims

1. A needle loader, comprising: a housing; a movable arm disposed within the housing, wherein the movable arm is slidably disposed along a first direction along a length of a cavity defined within the housing; and a needle mount defined on the movable arm to releasably attach to at least a portion of a suturing needle, wherein the needle mount and the at least a portion of the suturing needle are configured to deflect within the cavity along a second direction that is different from the first direction while the needle mount and the at least a portion of the suturing needle slide along the first direction from a first position to a second position, wherein the needle mount and the at least a portion of the suturing needle are spaced apart from a needle track of a suturing device in the first position, and wherein the needle mount and the at least a portion of the suturing needle are adjacent the needle track of the suturing device in the second position.
2. The needle loader of claim 1, further comprising at least a portion of a suturing needle releasably attached to the needle mount.
3. The needle loader of claim 2, wherein the at least a portion of the suturing needle comprises at least a portion of an arced suturing needle, and further wherein the at least a portion of the arced suturing needle is held in place and attached to the needle mount by way of an interference fit.
4. The needle loader of claim 1, wherein the needle mount is configured to lock in position after the needle mount has reached the second position.
5. The needle loader of claim 1, wherein the needle mount intersects the needle track of the suturing device when the needle mount and the at least a portion of the suturing needle have reached the second position, and further wherein the needle loader further comprises a spring that is configured to urge against motion of the needle mount as the needle mount moves from the first position toward the second position.
6. The needle loader of claim 1, wherein the movable arm includes a cammed surface that is configured to slide over a distal portion of the suturing device as the needle mount and the at least a portion of the suturing needle approaches the second position.
7. The needle loader of claim 1, wherein movement of the needle mount and the at least a portion of the suturing needle to the second position activates an indicator confirming that the needle mount and the at least a portion of the suturing needle have reached the second position.
8. The needle loader of claim 7 wherein the indicator confirms that the needle mount at the least a portion of the suturing needle have reached the second position by displaying visual indicia.
9. The needle loader of claim 1, further comprising a protrusion that extends into a path of the suturing device to be loaded with the at least a portion of the suturing needle from the needle loader, wherein the protrusion is configured to cause the needle track of the suturing device to be exposed as the suturing device is advanced along the path, and further wherein the suturing device includes a retractable cover to cover the needle track, and still further wherein the protrusion is

configured to retract the retractable cover of the suturing device to expose the needle track.

10. The needle loader of claim 1, wherein the second direction is non-linear.

11. The needle loader of claim 1, wherein the movable arm bends along the second direction while the movable arm, the needle mount, and the at least a portion of the suturing needle slide from the first position to the second position, and further wherein the first direction is linear.

12. The needle loader of claim 1, wherein the second direction is oblique or orthogonal with respect to the first direction.

13. A device for loading at least a portion of a suturing needle into a suturing device, comprising: a mount defining at least one indentation therein, the indentation being configured to receive at least a portion of the suturing needle therein in an interference fit, wherein the mount is configured to slide along a first direction from a first position wherein the mount and the at least a portion of the suturing needle is out of contact with a needle track of a suturing device, to a second position wherein the mount is adjacent the needle track of the suturing device and the at least a portion of the suturing needle is deposited into the needle track of the suturing device; and an elongate shaft extending from the mount, wherein the elongate shaft is configured to slide with the mount along a portion of the suturing device along the first direction as the mount and the at least a portion of the suturing needle is advanced from the first position to the second position, and further wherein the elongate shaft, the mount, and the at least a portion of the suturing needle are configured to deflect along a second direction that is different from the first direction as the mount and the at least a portion of the suturing needle are slid from the first position to the second position.

14. The device of claim 13, wherein the mount defines at least one guide boss thereon configured to be received by a guide track.

15. The device of claim 13, wherein the at least a portion of the suturing needle includes a sharpened leading end.

16. The device of claim 13, wherein at least a portion of the elongate shaft comprises a laterally deformable body, wherein the laterally deformable body is configured to be bent from a first shape into a second shape as the mount is advanced from the first position to the second position.

17. The device of claim 13, wherein the mount includes a magnet and the at least a portion of the suturing needle includes ferromagnetic material, and further wherein the magnet is configured to hold the at least a portion of the suturing needle in place until the at least a portion of the suturing needle is deposited in the needle track of the suturing device.

18. The device of claim 13, wherein the device further comprises an indicator to indicate that the at least a portion of the suturing needle has been loaded into the needle track of the suturing device.

19. The device of claim 18, wherein the indicator includes a tactile indicator.

20. The device of claim 13, wherein the second direction is non-linear.

21. The device of claim 13, wherein the elongate shaft bends along the second direction while the elongate shaft, mount and the at least a portion of the suturing needle slide from the first position to the second position, and further wherein the first direction is linear.

22. The device of claim 13, wherein the second direction is oblique or orthogonal with respect to the first direction.
